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Name: _____

November 10, 2019 Sunday

Department/School: _____

SID: _____

University Physics II

Midterm Examination

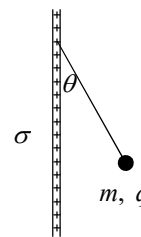
Kuang Yaming Honors School, Nanjing University

Select five out of following six problems.

1. (20pts.) An infinite non-conducting vertical planar sheet is charged homogeneously with areal electric charge density σ .

(a) Find the electric field $\mathbf{E}$ everywhere, specifying the normal direction of the sheet as the direction of x -axis.

(b) A silk thread with one end suspended on the sheet and the other end attached to a small charged sphere of mass m makes an angle $\theta = \pi/6$ with the sheet, as shown in the figure. What is the electric charge q on the sphere?



2.(20pts.) An insulating rod of length l is positioned on the z -axis, from $z = -l/2$ to $z = l/2$, and it carries an electric charge of linear charge density λ .

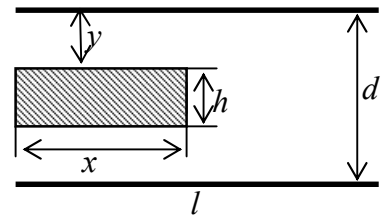
(a) Calculate the electric potential $V = V(z)$ on the z -axis and within the region $z > l/2$ and $z < -l/2$. The potential at infinity is set to zero.

(b) Calculate the electric potential $V = V(x, y, z)$ everywhere, except on the z -axis. The potential at infinity is set to zero.

(c) Calculate the electric field $\mathbf{E} = \mathbf{E}(x, y, z)$ everywhere, except on the z -axis.

(Hint: $\int \frac{d\xi}{\sqrt{\xi^2 + 1}} = \ln|\xi + \sqrt{\xi^2 + 1}| + C$)

3. (20pts.) As shown in the figure, a copper slab of thickness h is parallelly inserted into a square parallel-plate capacitor of separation d and side length l . The distance between the upper surface of the slab and the upper plate of the capacitor is y . The slab is rectangular of length l and width x ($x < l$).

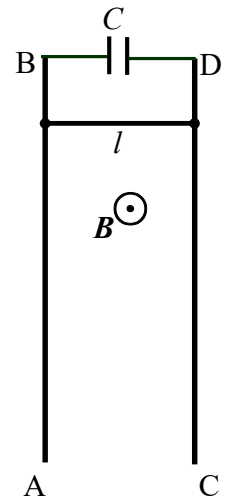


(a) What is capacitance C of the system?

(b) If the charge on the capacitor is q , what is the electric potential energy U of the system?

(c) What is the horizontal force f_1 exerted on the slab by the electric field? And what is the vertical force f_2 ? Suppose the charge on the capacitor is fixed.

4. (20pts.) A conducting rod of length l and mass m can slide without friction along two sufficiently long vertical parallel conducting rails AB and CD that are connected each other through a capacitor of capacitance C at the upper ends, as shown in the figure. Perpendicular to the plane of the rails, a uniform magnetic field of induction $\mathbf{B}$ is applied. Assume that the conducting rod and the conducting rails are made of ideal conductors of zero resistance and there is also no contact resistance. Initially, the rod is at rest and the capacitor is uncharged.

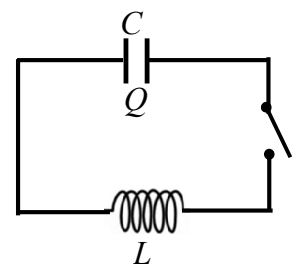


(a) Find the voltage V across the capacitor as a function of h , the lowered height of the rod from its initial position.

(b) If the capacitor is replaced by a resistor of resistance R , find the voltage V across the resistor as a function of time t .

(c) Discuss the meaning of the result (b) in the limit $t \rightarrow +\infty$.

5. (20pts.) In the circuit shown in the accompanying figure, the capacitor of capacitance C is initially charged with electric charge Q . The switch is closed at time $t = 0$ and the electric current begins to flow through the inductor of inductance L .



(a) Write down the equation of oscillation for the charge q on the capacitor. Assume the resistance in the circuit is negligible.

(b) Find the solution of charge q on the capacitor as a function of time t .

(c) Find the total electromagnetic energy U stored in the capacitor and inductor at time t .

6. (20pts.) An electromagnetic wave travels in free space (i.e., in the vacuum), and its magnetic induction $\mathbf{B}$ takes the form

$$\mathbf{B} = (b_1 \mathbf{e}_x + b_2 \mathbf{e}_y) \cos(kz - \omega t)$$

where b_1 , b_2 , ω and k are all real constants, $\omega = ck$ with c being the speed of light in the vacuum, $\mathbf{e}_x$ and $\mathbf{e}_y$ are the unit vectors along the x - and y - axis, respectively.

(a) Find, by applying Maxwell equations, the electric field $\mathbf{E}$ and the displacement current density $\mathbf{j}_D$.

(b) What is the Poynting vector $\mathbf{S}$ of this wave? What is the average Poynting vector $\langle \mathbf{S} \rangle$ over one period of time?